The multiplier of a long-root collapse

Motegi's type B/C weight functions degenerate on $w_a w_b = 1$ by a universal scalar ρ; the Laurent poles of the structure constants are measured by $\rho - 1$, and $\rho - 1$ carries exactly one factor of q

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Abstract

This continues `proofs/2026-10-16-Q488-korepin-locus.tex`, henceforth [Q488], whose Theorem `main` proved one direction of the Laurent criterion $|\nu| - \max(|\lambda|, |\mu|) \le 1$ for the structure constants $c^{\nu}_{\lambda\mu}$ of Motegi's type C fixed-point algebra (arXiv:2610.15344), and left the converse as Gap `nonvanishing` ("jump ≥ 2 implies a pole *actually occurs*").

The session was gated on reproducing Rick's counter-measurement that at $q = 0$ every structure constant is Laurent. It reproduces, and it is *vacuous on exactly the triples it was aimed at*: $\operatorname{ord}_q c^{\nu}_{\lambda\mu} = 1$ in 25/25 pole records at $N = 3$, so the non-Laurent constants are precisely the ones that *vanish* identically at $q = 0$. Zero is Laurent.

Chasing where that factor of q lives produced the mechanism. On a long-root divisor $H_{ab} = \{w_a w_b = 1\}$ every factor of Motegi's branching formula with one leg in $\{a, b\}$ pairs with its own inverse, so the branch index and the complement both drop out and

$$\widetilde{A}_{\mathbf{M},\mathbf{I}} \equiv \rho \cdot \widetilde{A}_{\mathbf{M}\setminus\{a,b\},\mathbf{I}} \pmod{w_a w_b - 1}, \qquad \rho = \frac{(1-q)^2}{(1-qw_a^2)(1-qw_a^{-2})},$$

for *every* $\mathbf{I}$, with ρ independent of $\mathbf{M}$, $\mathbf{I}$, the branch index and N (Theorem 5; types C and B alike). So a long-root divisor *is* a collapse — twisted by a scalar. Basis elements collapse with multiplier ρ and their products with ρ^2, and the entire pole phenomenon is measured by

$$\rho - 1 = \frac{q\,(w_a - w_a^{-1})^2}{(1-qw_a^2)(1-qw_a^{-2})} \neq 0.$$

This is simultaneously the proof that poles occur and the explanation of the $q = 0$ measurement: the numerator carries exactly one factor of q, so every long-root residue has $\operatorname{ord}_q = 1$ (predicted, then matched 25/25), and at $q = 0$ the twist trivialises and the long-root divisor becomes an honest collision divisor.

Four theorems follow. The "$\Leftarrow$" half of the exact-pole-divisor conjecture of the previous paper is proved outright (Theorem 10). Every long-root pole is simple (Theorem 13). For ν *minimal*

with the pole condition, the created pole is real, with a closed-form residue (Theorem 15) — this closes Gap `nonvanishing` on that stratum, modulo one named non-vanishing hypothesis. And the residues obey an exact recursion (Theorem 12).

I report the scope honestly: the minimal-ν stratum is 1/1 of the jump-≥ 2 triples at $N = 2$, 12/13 at $N = 3$, but only 99/127 at $N = 4$, 690/1136 at $N = 5$ and 4335/9748 at $N = 6$. **Coverage decreases with** N, so this is a partial closure, not a closure, and the residual gap `propagation` is the bulk of the problem rather than an edge case. It is now a precisely stated non-cancellation in the recursion of Theorem 12, rather than "the absence of conspiratorial cancellation" among branch terms.

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1 The gate, and what the $q = 0$ measurement says

The session brief armed a thirty-minute gate at position zero, to reproduce a counter-measurement before building anything. It ran. It did not return any of the three pre-registered outcomes.

1.1 What was measured

Rick's review of Q484 (UID 813, 2026-10-16 01:05) claims, as his M2: *at $q = 0$ every structure constant is Laurent* — type C $N = 2$: 30/30; type C $N = 3$: 195/195; type B $N = 2$: 30/30 — and adds "my witness carries a factor q and vanishes at $q = 0$".

Reproduced on my own engine (`q484_engine.py`, brute-force summation over S_n from Motegi's symmetrisation formula L474–L487, which never uses the branching formula (4) below):

	nonzero constants	pole records (triple, D)	pole triples	
$N = 2$, type C	20	1	1	
$N = 3$, type C	115	25	13	(12 at jump 2, 1 at jump 3)

These agree with the previous paper's own verification section, which records "jump 2 gives 0 Laurent and $12+1$ poles; jump 3 gives 0 Laurent and 1 pole". The decisive columns are new:

$$\operatorname{ord}_q c^{\nu}_{\lambda\mu} = \operatorname{ord}_q \operatorname{Res}_H c^{\nu}_{\lambda\mu} = 1 \quad \text{in } 25/25 \text{ pole records, every pole simple.} \tag{1}$$

Remark 1 (a carried number, corrected). The brief asserted that the exact-pole-divisor conjecture "predicts 19 poles at $N = 3$". The figure 19 occurs neither in the previous paper nor in the measurement; the correct figures are 13 triples and 25 (triple, divisor) records. It was carried, not re-derived.

1.2 Why Rick's M2 is true and carries no information

Here is the whole of it, on the unique $N = 2$ witness. With $P \in \mathbb{Z}[q, w_1, w_2]$ and $P(0, \mathbf{w}) = w_1 w_2$,

$$c^{\{1,2\}}_{\emptyset\emptyset} = \frac{-q\, w_1 w_2\, (q-1)^2\, P(q, \mathbf{w})}{w_1 w_2 - 1}, \qquad \operatorname{Res}_{w_1 w_2 = 1} c^{\{1,2\}}_{\emptyset\emptyset} = -q(q-1)^4.$$

The factor q sits in the *whole* constant, not only in the residue. So $c^{\{1,2\}}_{\emptyset\emptyset}\big|_{q=0} = 0$, and 0 is a Laurent polynomial. The verdict "Laurent at $q = 0$" is satisfied by vanishing, not by cancellation of the pole. Normalising by the leading power of q keeps it:

$$\lim_{q \to 0}\; q^{-1} c^{\{1,2\}}_{\emptyset\emptyset} = \frac{-w_1^2 w_2^2}{w_1 w_2 - 1}.$$

By (1) this is uniform: in every one of the 25 pole records at $N = 3$, $\operatorname{ord}_q c = 1$, hence $c|_{q=0} = 0$.

Remark 2 (the brief's pre-registered inference does not follow). The brief pre-registered: "(0A) It reproduces. Then the converse as written in the tex is FALSE, and the first act of the session is to amend Gap `nonvanishing` *in the paper* to carry a genericity hypothesis in q." That instruction is withdrawn. The structure constants are elements of $\mathbb{F} = \mathbb{C}(q)(w_1, \ldots, w_N)$, in which q is an *indeterminate*; "c has a pole on H" is a statement about the denominator of an element of $\mathbb{F}$, already generic in q, and specialisation $q \mapsto 0$ does not commute with it. Gap `nonvanishing` as written is therefore untouched by M2 and needs *no* genericity hypothesis. A counter-measurement can refute an inference drawn from a claim without refuting the claim.

What the gate does deliver is the asset: the residue carries *exactly one* factor of q. That is a quantitative fact, and §3 turns out to predict it.

2 Setup

Notation is that of [Q488], and all facts cited to Motegi are from arXiv:2610.15344 with the line numbers of its e-print source as recorded there. Briefly: $\mathbb{F} = \mathbb{C}(q)(w_1, \ldots, w_N)$; for $\mathbf{M}, \mathbf{I} \subseteq [N]$, $\widetilde{A}_{\mathbf{M},\mathbf{I}} = \widetilde{WC}_{N,n,|\mathbf{I}|}(\mathbf{w}_{\mathbf{M}}|\mathbf{w}|\mathbf{I})$; the $\varphi_{\mathbf{I}} : \mathbf{M} \mapsto \widetilde{A}_{\mathbf{M},\mathbf{I}}$ form a basis of the algebra of $\mathbb{F}$-valued functions on the 2^N specialisation points, with

$$\widetilde{A}_{\mathbf{M},\lambda} \widetilde{A}_{\mathbf{M},\mu} = \sum_{\nu} c^{\nu}_{\lambda\mu} \widetilde{A}_{\mathbf{M},\nu} \qquad \text{for every } \mathbf{M}, \tag{2}$$

and, along a linear extension $\prec$ of Motegi's order,

$$\widetilde{A}_{\nu,\nu}\, c^{\nu}_{\lambda\mu} = \widetilde{A}_{\nu,\lambda} \widetilde{A}_{\nu,\mu} - \sum_{\kappa \prec \nu} c^{\kappa}_{\lambda\mu} \widetilde{A}_{\nu,\kappa}. \tag{3}$$

Facts imported from [Q488] and used below, with their status there:

- **(T)** triangularity, $\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ unless $\mathbf{M} \geq \mathbf{I}$ (Motegi Thm 2.6(i); *proved*);
- **(R)** regularity, $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ is a Laurent polynomial (Motegi; *proved*, deep-read);
- **(D)** the diagonal product formula `normalizationWC` (*proved*), from which $\operatorname{ord}_{H_{ab}} \widetilde{A}_{\mathbf{I},\mathbf{I}} = 1$ if $\{a,b\} \subseteq \mathbf{I}$ and $= 0$ otherwise;
- **(S)** the support bound: $c^{\nu}_{\lambda\mu} \neq 0$ implies $\nu \geq \lambda$ and $\nu \geq \mu$ (Q484 Theorem A; *proved*);
- **(A)** Lemma A of [Q488]: if $\{a,b\} \subseteq \mathbf{M}$ and $\mathbf{M} \setminus \{a,b\} \not\geq \mathbf{I}$ then $\operatorname{ord}_{H_{ab}} \widetilde{A}_{\mathbf{M},\mathbf{I}} \geq 1$ (*proved* from the branching formula);
- **(B)** the branching formula `eq:typeC-one-branch-explicit` (L3203–L3236), reproduced as (4) below (*proved*, Motegi's).

Throughout, $a < b$ are fixed, $H := H_{ab} = \{w_a w_b = 1\}$, and $t := w_a w_b - 1$ is a local parameter for H. Write $\mathcal{O}$ for the local ring of $\mathbb{F}$ at H (a discrete valuation ring with valuation $\operatorname{ord} = \operatorname{ord}_H$ and residue field $\mathcal{K}$, the function field of H), and $f \mapsto f|_H$ for reduction $\mathcal{O} \to \mathcal{K}$. By (R) every $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ lies in $\mathcal{O}$.

It is convenient to restate Motegi's order in the form that makes its two elementary properties one-liners.

Lemma 3 (the order, restated)**.** *For $\mathbf{M},\mathbf{I} \subseteq [N]$, Motegi's $\mathbf{M} \geq \mathbf{I}$ holds if and only if $|\mathbf{M}| \geq |\mathbf{I}|$ and, for every $1 \leq j \leq |\mathbf{I}|$, the j-th largest element of $\mathbf{M}$ is $\geq$ the j-th largest element of $\mathbf{I}$. Consequently:*

(i) $\geq$ is transitive;

(ii) if $\mathbf{M}' \subseteq \mathbf{M}$ then $\mathbf{M}' \geq \mathbf{I}$ implies $\mathbf{M} \geq \mathbf{I}$.

Proof. Writing $\mathbf{M} = \{m_1 < \cdots < m_s\}$ and $\mathbf{I} = \{i_1 < \cdots < i_r\}$, the element m_{s-r+j} is the $(r-j+1)$-st largest of $\mathbf{M}$ and i_j is the $(r-j+1)$-st largest of $\mathbf{I}$, so the restated condition is exactly $m_{s-r+j} \geq i_j$ for all j, which is Motegi's (L640–L650). (i): if $\mathbf{M} \geq \mathbf{I} \geq \mathbf{J}$ then for $j \leq |\mathbf{J}|$ the j-th largest of $\mathbf{M}$ is $\geq$ the j-th largest of $\mathbf{I}$, which is $\geq$ the j-th largest of $\mathbf{J}$; and $|\mathbf{M}| \geq |\mathbf{I}| \geq |\mathbf{J}|$. (ii): for $\mathbf{M}' \subseteq \mathbf{M}$ the j-th largest element of $\mathbf{M}'$ is $\leq$ the j-th largest element of $\mathbf{M}$, for every $j \leq |\mathbf{M}'|$. □

3 The twisted collapse

Motegi's branching formula is the following. For $\mathbf{K} \subseteq \mathbf{M}$ with $|\mathbf{K}| = r = |\mathbf{I}|$ and $\mathbf{L} := \mathbf{M} \setminus \mathbf{K}$, he writes $\widetilde{A}_{\mathbf{M},\mathbf{I}} = \sum_{\mathbf{K}} T_{\mathbf{K}}$ with (indices of the inner products renamed to α, β, to free a, b for the long root)

$$T_{\mathbf{K}} = (1-q)^{|\mathbf{L}|} \cdot \Pi_1 \cdot \Phi_{\mathbf{K},\mathbf{I}} \cdot \Pi_3 \cdot \Pi_4 \cdot \Pi_5, \tag{4}$$

$$\Pi_1 = \prod_{\substack{j \leq k \\ j,k \in \mathbf{M}^{\mathrm{c}}}} (1 - q w_j w_k), \qquad \Phi_{\mathbf{K},\mathbf{I}} = \frac{W_{N,r}(w_{k_1}, \ldots, w_{k_r} | \mathbf{w}|\mathbf{I})}{\prod_{\alpha,\beta=1}^{r} (1 - q w_{k_\alpha}^{-1} w_{k_\beta})} \prod_{1 \leq \alpha \leq \beta \leq r} \bigl(1 - w_{k_\alpha}^{-1} w_{k_\beta}^{-1}\bigr),$$

$$\Pi_3 = \frac{\prod_{\alpha=1}^{r} \prod_{l \in \mathbf{L}} \bigl(1 - w_{k_\alpha}^{-1} w_l^{-1}\bigr)}{\prod_{\alpha=1}^{r} \prod_{l \in \mathbf{L}} \bigl(1 - w_{k_\alpha}^{-1} w_l\bigr)}, \qquad \Pi_4 = \prod_{\{l,l'\} \subseteq \mathbf{L}} \bigl(1 - q w_l^{-1} w_{l'}^{-1}\bigr),$$

$$\Pi_5 = \prod_{l \in \mathbf{L}} \prod_{j \in \mathbf{M}^{\mathrm{c}}} \bigl(1 - q w_l^{-1} w_j\bigr).$$

(The factor Π_4 is symmetric in l, l', both legs being inverted, so indexing it by unordered pairs is legitimate.)

Lemma 4 (the slot structure of (4))**.** *In* (4) *the label* $\mathbf{I}$ *occurs only inside* $\Phi_{\mathbf{K},\mathbf{I}}$*, and* $\Phi_{\mathbf{K},\mathbf{I}}$ *depends only on the pair* $(\mathbf{K},\mathbf{I})$ *— not on* $\mathbf{L}$ *and not on* $\mathbf{M}^{\mathrm{c}}$*. The remaining factors* $\Psi_{\mathbf{K},\mathbf{L},\mathbf{M}^{\mathrm{c}}} := (1-q)^{|\mathbf{L}|}\Pi_1\Pi_3\Pi_4\Pi_5$ *do not involve* $\mathbf{I}$ *at all.*

Proof. Read off (4): $\mathbf{I}$ appears exactly once, as the label of $W_{N,r}$; and the three factors constituting Φ involve only $w_{k_1},\dots,w_{k_r}$, the global alphabet $\mathbf{w}$, and $\mathbf{I}$. □

This is the structural half of Gap `q` of [Q488] — there proved only for the normalisation $\mathcal{E}^C$ — promoted to the whole formula. It is what makes the next theorem uniform in $\mathbf{I}$.

Theorem 5 (twisted collapse on a long root)**.** *Let* $a<b$ *with* $\{a,b\}\subseteq\mathbf{M}$*, and* $\mathbf{M}' := \mathbf{M}\setminus\{a,b\}$*. Then for* every $\mathbf{I}\subseteq[N]$*,*

$$\widetilde{A}_{\mathbf{M},\mathbf{I}} \;\equiv\; \rho\cdot\widetilde{A}_{\mathbf{M}',\mathbf{I}} \pmod{w_aw_b-1}, \qquad \rho := \frac{(1-q)^2}{(1-qw_a^2)\,(1-qw_a^{-2})}, \tag{5}$$

i.e. $\operatorname{ord}_H\big(\widetilde{A}_{\mathbf{M},\mathbf{I}}-\rho\widetilde{A}_{\mathbf{M}',\mathbf{I}}\big)\ge 1$*. The multiplier* ρ *depends on neither* $\mathbf{M}$*, nor* $\mathbf{I}$*, nor the branch index, nor* N*; it is symmetric in* $a\leftrightarrow b$*, since* $w_b^2=w_a^{-2}$ *on* H*. The same statement and the same* ρ *hold in type* B*.*

Proof. By (A) read through (4) — this is Lemma `branchord` of [Q488] — every $T_{\mathbf{K}}$ with $\mathbf{K}\cap\{a,b\}\neq\emptyset$ has $\operatorname{ord}_H T_{\mathbf{K}}\ge 1$: such a $\mathbf{K}$ contributes a factor $\big(1-w_{k_\alpha}^{-1}w_{k_\beta}^{-1}\big)$ with $\{k_\alpha,k_\beta\}=\{a,b\}$ from Φ, or a factor $\big(1-w_{k_\alpha}^{-1}w_l^{-1}\big)$ with $\{k_\alpha,l\}=\{a,b\}$ from Π_3, and both vanish on H. Also, by [Q488] §4, no denominator of (4) vanishes on a long-root divisor: the denominators are $\prod(1-qw_{k_\alpha}^{-1}w_{k_\beta})$ and $\prod(1-w_{k_\alpha}^{-1}w_l)$, of the q-shifted and type A families, and $W_{N,r}$ is a polynomial. Hence every $T_{\mathbf{K}}$ lies in $\mathcal{O}$ and

$$\widetilde{A}_{\mathbf{M},\mathbf{I}}\big|_H = \sum_{\substack{\mathbf{K}\subseteq\mathbf{M}',\ |\mathbf{K}|=r\\ \mathbf{K}\supseteq\mathbf{I}}} T_{\mathbf{K}}\big|_H, \tag{6}$$

the admissibility constraint $\mathbf{K}\supseteq\mathbf{I}$ being Motegi's (Lemma `adm` of [Q488]). The index set of (6) is exactly the index set of the branching formula for $\mathbf{M}'$, whose terms we write $T'_{\mathbf{K}}$.

Fix such a $\mathbf{K}$ and put $\mathbf{L}' := \mathbf{M}'\setminus\mathbf{K}$, so that for $\mathbf{M}$ we have $\mathbf{L}=\mathbf{L}'\sqcup\{a,b\}$ and $\mathbf{M}^{\mathrm{c}}$, while for $\mathbf{M}'$ we have $\mathbf{L}'$ and $\mathbf{M}'^{\mathrm{c}}=\mathbf{M}^{\mathrm{c}}\sqcup\{a,b\}$. By Lemma 4 the factor $\Phi_{\mathbf{K},\mathbf{I}}$ is common to $T_{\mathbf{K}}$ and $T'_{\mathbf{K}}$, so

$$\frac{T_{\mathbf{K}}}{T'_{\mathbf{K}}} = \frac{\Psi_{\mathbf{K},\mathbf{L}'\sqcup\{a,b\},\mathbf{M}^{\mathrm{c}}}}{\Psi_{\mathbf{K},\mathbf{L}',\mathbf{M}^{\mathrm{c}}\sqcup\{a,b\}}},$$

which is independent of $\mathbf{I}$. Factor by factor, the quotient is:

source	contribution to $T_{\mathbf{K}}/T'_{\mathbf{K}}$
$(1-q)^{\lvert\mathbf{L}\rvert}$	$(1-q)^2$
Π_1	$\dfrac{1}{(1-qw_a^2)(1-qw_b^2)(1-qw_aw_b)\prod_{j\in\mathbf{M}^{\mathrm{c}}}(1-qw_aw_j)(1-qw_bw_j)}$
Π_3	$\displaystyle\prod_{k\in\mathbf{K}} h(k),\quad h(k)=\frac{(1-w_k^{-1}w_a^{-1})(1-w_k^{-1}w_b^{-1})}{(1-w_k^{-1}w_a)(1-w_k^{-1}w_b)}$
Π_4	$(1-qw_a^{-1}w_b^{-1})\displaystyle\prod_{l\in\mathbf{L}'}(1-qw_l^{-1}w_a^{-1})(1-qw_l^{-1}w_b^{-1})$
Π_5	$\dfrac{\prod_{j\in\mathbf{M}^{\mathrm{c}}}(1-qw_a^{-1}w_j)(1-qw_b^{-1}w_j)}{\prod_{l\in\mathbf{L}'}(1-qw_l^{-1}w_a)(1-qw_l^{-1}w_b)}$

Now restrict to H, where $w_a^{-1} = w_b$ and $w_b^{-1} = w_a$. Every factor above with *one* leg in $\{a, b\}$ is accompanied by its own image under $w_a \leftrightarrow w_b$, and the four cancellations are:

- $h(k)\big|_H = \dfrac{(1 - w_k^{-1}w_b)(1 - w_k^{-1}w_a)}{(1 - w_k^{-1}w_a)(1 - w_k^{-1}w_b)} = 1$ for every k: *the branch index drops out*;
- the $\mathbf{L}'$-product of Π_4 against the $\mathbf{L}'$-denominator of Π_5: $(1 - qw_l^{-1}w_a^{-1})(1 - qw_l^{-1}w_b^{-1})\big|_H = (1 - qw_l^{-1}w_b)(1 - qw_l^{-1}w_a)$: $\mathbf{L}'$ *drops out*;
- the $\mathbf{M}^{\mathrm{c}}$-numerator of Π_5 against the $\mathbf{M}^{\mathrm{c}}$-denominator of Π_1: $(1 - qw_a^{-1}w_j)(1 - qw_b^{-1}w_j)\big|_H = (1 - qw_bw_j)(1 - qw_aw_j)$: $\mathbf{M}^{\mathrm{c}}$ *drops out*;
- $(1 - qw_a^{-1}w_b^{-1})\big|_H = 1 - q = (1 - qw_aw_b)\big|_H$, since $w_a^{-1}w_b^{-1} = (w_aw_b)^{-1} = 1$ on H.

What survives is the one factor of Π_1 with *both* legs in $\{a, b\}$, against the leading $(1 - q)^2$:

$$\frac{T_{\mathbf{K}}}{T'_{\mathbf{K}}}\bigg|_H = \frac{(1 - q)^2}{(1 - qw_a^2)(1 - qw_b^2)}\bigg|_H = \frac{(1 - q)^2}{(1 - qw_a^2)(1 - qw_a^{-2})} = \rho.$$

None of the factors evaluated above is identically 0 on H — the only candidates were $(1 - qw_aw_b)$ and $(1 - qw_a^{-1}w_b^{-1})$, both equal to $1 - q \neq 0$ there, q being an indeterminate — so this is an identity of nonzero elements of $\mathcal{K}$. Summing (6) over the common index set gives $\widetilde{A}_{\mathbf{M},\mathbf{I}}|_H = \rho \sum_{\mathbf{K}} T'_{\mathbf{K}}|_H = \rho\, \widetilde{A}_{\mathbf{M}',\mathbf{I}}|_H$.

For type B: the type B specialisation differs from type C by $\beta = 1, \gamma = -1$ and the prefactor $\prod_k (1 + u_k^{-1})^{-1}$, neither of which contributes a factor vanishing on a long-root divisor, and the cross-sector branching formula has the same shape. The claim is verified independently in §9. □

Remark 6 (what this corrects, and what it does not). [Q488] opens its §6 with "The long-root divisors $w_aw_b = 1$ are *not* collision divisors: setting $w_b = w_a^{-1}$ does not make $\mathbf{w}_{\mathbf{M}}$ coincide with any other specialisation point … This is the whole asymmetry, and it is where poles live." The conclusion stands and the reason is incomplete. There *is* a collapse of the row $\mathbf{M}$ onto the row $\mathbf{M} \setminus \{a, b\}$, uniform in $\mathbf{I}$; what distinguishes a long root from a collision is not the absence of a collapse but the presence of a *multiplier* $\rho \neq 1$. Poles are measured by $\rho - 1$, and that is the content of the next section. The specialisation points do not collide; the *rows* do, projectively.

4 The multiplier is the obstruction

Proposition 7. $\rho - 1 = \dfrac{q\,(w_a - w_a^{-1})^2}{(1 - qw_a^2)(1 - qw_a^{-2})}$. *In particular* $\operatorname{ord}_H \rho = \operatorname{ord}_H(\rho - 1) = 0$, *and* $\rho \neq 1$ *in* $\mathbb{F}$.

Proof. $(1 - q)^2 - (1 - qw_a^2)(1 - qw_a^{-2}) = (1 - 2q + q^2) - (1 - qw_a^{-2} - qw_a^2 + q^2) = q\big(w_a^2 - 2 + w_a^{-2}\big) = q(w_a - w_a^{-1})^2$. Neither numerator nor denominator involves w_b, so neither is divisible by $w_aw_b - 1$. □

Three readings of that one line.

(i) It forces the poles. Basis elements collapse with multiplier ρ; their pointwise products collapse with ρ^2. Indeed, for $\{a,b\} \subseteq \mathbf{M}$ and $\mathbf{M}' = \mathbf{M} \setminus \{a,b\}$, Theorem 5 gives

$$\big(\widetilde{A}_{\mathbf{M},\lambda}\widetilde{A}_{\mathbf{M},\mu}\big)\big|_H = \rho^2\big(\widetilde{A}_{\mathbf{M}',\lambda}\widetilde{A}_{\mathbf{M}',\mu}\big)\big|_H. \tag{7}$$

If every $c^\nu_{\lambda\mu}$ were in $\mathcal{O}$, then reducing (2) at $\mathbf{M}$ and at $\mathbf{M}'$ and using Theorem 5 column by column would give $\big(\widetilde{A}_{\mathbf{M},\lambda}\widetilde{A}_{\mathbf{M},\mu}\big)|_H = \sum_\nu c^\nu|_H \widetilde{A}_{\mathbf{M},\nu}|_H = \rho\sum_\nu c^\nu|_H \widetilde{A}_{\mathbf{M}',\nu}|_H = \rho\big(\widetilde{A}_{\mathbf{M}',\lambda}\widetilde{A}_{\mathbf{M}',\mu}\big)|_H$, multiplier ρ, contradicting (7) unless $\rho(\rho-1)\big(\widetilde{A}_{\mathbf{M}',\lambda}\widetilde{A}_{\mathbf{M}',\mu}\big)|_H = 0$. Since $\rho(\rho-1) \neq 0$ by Proposition 7, we obtain at once:

Corollary 8 (the obstruction, qualitative)**.** *If there exists $\mathbf{M}' \subseteq [N]$ with $\mathbf{M}' \cap \{a,b\} = \emptyset$ and $\widetilde{A}_{\mathbf{M}',\lambda}|_H \neq 0 \neq \widetilde{A}_{\mathbf{M}',\mu}|_H$, then not every $c^\nu_{\lambda\mu}$ is regular on H_{ab}; some $c^\nu_{\lambda\mu}$ has a pole there.*

(ii) It predicts $\operatorname{ord}_q = 1$**.** The numerator of $\rho - 1$ is $q(w_a - w_a^{-1})^2$: exactly one factor of q, and the denominator of $\rho - 1$ is a unit at $q = 0$. Every residue computed in §7 is proportional to $\rho - 1$, so

$$\operatorname{ord}_q \operatorname{Res}_H c^\nu_{\lambda\mu} = 1.$$

This was derived from (4) and matched afterwards against the 25 pole records of (1): a prediction confirmed, not a fit.

(iii) It locates the exceptional locus in q exactly. $\rho - 1$ vanishes precisely on $\{q = 0\}$ and on $\{w_a^2 = 1\}$. The second is the short-root divisor; its intersection with H_{ab} has codimension 2 and is invisible to a statement about divisors. So:

Corollary 9 (Rick's M2, derived)**.** $\rho|_{q=0} = 1$*. At $q = 0$ the twist is trivial, so the long-root divisor becomes a collision divisor in the sense of [Q488] §5, with collapse map $\mathbf{M} \mapsto \mathbf{M} \setminus \{a,b\}$ — and Theorem* `collision` *of [Q488] then forbids a pole there, for every (λ,μ,ν). The exceptional locus of the converse direction, as a subset of the q-line, is exactly $\{0\}$ (together with $q = \infty$, where $\rho \to 1$ as well).*

This is the sense in which the counter-measurement handed over the mechanism: asking *where* the factor q lives answers *why* there is a pole at all.

5 The easy direction of the exact pole divisor

[Q488] states the exact pole divisor as a conjecture, verified 20/20 and 115/115. One of its two directions is now a theorem, and it is needed in §7.

Theorem 10 ($\Leftarrow$ of the exact-pole-divisor conjecture)**.** *Fix $a < b$ and λ,μ. Say that κ* brackets *(a,b) if $\{a,b\} \subseteq \kappa$ and $\kappa \setminus \{a,b\} \geq \lambda$ and $\kappa \setminus \{a,b\} \geq \mu$. If no κ with $\nu \geq \kappa$, $\kappa \geq \lambda$, $\kappa \geq \mu$ brackets (a,b), then $\operatorname{ord}_{H_{ab}} c^\nu_{\lambda\mu} \geq 0$.*

Proof. Induction along $\prec$. Note first that the hypothesis is inherited: if $\nu \geq \kappa$ then $\{\kappa' \leq \kappa\} \subseteq \{\kappa' \leq \nu\}$ by Lemma 3(i), so the hypothesis for ν implies it for κ. In (3) only κ with $\widetilde{A}_{\nu,\kappa} \neq 0$ and $c^\kappa \neq 0$ contribute, i.e. (by (T) and (S)) only κ with $\nu \geq \kappa$ and $\kappa \geq \lambda$, $\kappa \geq \mu$; the inductive hypothesis is available for each.

Case $\{a,b\} \not\subseteq \nu$. By (D), $\operatorname{ord}_H \widetilde{A}_{\nu,\nu} = 0$. Every term on the right of (3) has $\operatorname{ord}_H \geq 0$, by (R) and induction. Hence $\operatorname{ord}_H c^\nu \geq 0$.

Case $\{a,b\} \subseteq \nu$. Put $\nu' := \nu \setminus \{a,b\}$. By (A) and (D), $\operatorname{ord}_H \widetilde{A}_{\nu,\nu} = 1$ exactly: ≥ 1 because $\nu' \not\geq \nu$ on cardinality grounds, and ≤ 1 because the only factor of the diagonal formula (D) vanishing on H is $(1 - w_a^{-1} w_b^{-1})$, which occurs once. It suffices that the right side of (3) have $\operatorname{ord}_H \geq 1$.

The hypothesis applied at $\kappa = \nu$ says ν does not bracket (a,b); since $\{a,b\} \subseteq \nu$, this means $\nu' \not\geq \lambda$ or $\nu' \not\geq \mu$ — say $\nu' \not\geq \lambda$. Then (A) gives $\operatorname{ord}_H \widetilde{A}_{\nu,\lambda} \geq 1$, and $\operatorname{ord}_H \widetilde{A}_{\nu,\mu} \geq 0$ by (R), so the first term has $\operatorname{ord}_H \geq 1$.

For a contributing κ: we have $\kappa \geq \lambda$. If $\nu' \geq \kappa$ then $\nu' \geq \lambda$ by transitivity (Lemma 3(i)), contradicting $\nu' \not\geq \lambda$. So $\nu' \not\geq \kappa$, and (A) gives $\operatorname{ord}_H \widetilde{A}_{\nu,\kappa} \geq 1$; with $\operatorname{ord}_H c^{\kappa} \geq 0$ by induction, that term has $\operatorname{ord}_H \geq 1$. □

Remark 11. Theorem 10 strictly generalises Theorem `main` of [Q488], which is the special case in which the combinatorial lemma `comb` of [Q488] makes the hypothesis hold simultaneously for every pair (a,b) — namely when $|\nu| - \max(|\lambda|, |\mu|) \leq 1$. The point of the generalisation is that it is available *divisor by divisor*, which is what the minimality argument of §7 consumes.

6 Every long-root pole is simple

Applying the collapse to (2) rather than to (3) gives an exact identity which does the bookkeeping of pole propagation for free. For $\{a,b\} \subseteq \mathbf{M}$, $\mathbf{M}' = \mathbf{M} \setminus \{a,b\}$ and any $\mathbf{I}$, put

$$u_{\mathbf{M},\mathbf{I}} := \frac{\widetilde{A}_{\mathbf{M},\mathbf{I}} - \rho\, \widetilde{A}_{\mathbf{M}',\mathbf{I}}}{t} \;\in\; \mathcal{O}, \tag{8}$$

which lies in $\mathcal{O}$ by Theorem 5.

Theorem 12 (the obstruction identity)**.** *For every* $\mathbf{M} \supseteq \{a,b\}$*, with* $\mathbf{M}' = \mathbf{M} \setminus \{a,b\}$*,*

$$\sum_{\nu} c^{\nu}_{\lambda\mu}\, u_{\mathbf{M},\nu} = \frac{\rho(\rho-1)\, \widetilde{A}_{\mathbf{M}',\lambda} \widetilde{A}_{\mathbf{M}',\mu}}{t} + \rho\big(\widetilde{A}_{\mathbf{M}',\lambda} u_{\mathbf{M},\mu} + \widetilde{A}_{\mathbf{M}',\mu} u_{\mathbf{M},\lambda}\big) + t\, u_{\mathbf{M},\lambda} u_{\mathbf{M},\mu}, \tag{9}$$

an identity in $\mathbb{F}$*, in which the last two terms lie in* $\mathcal{O}$*.*

Proof. Subtract ρ times (2) at $\mathbf{M}'$ from (2) at $\mathbf{M}$:

$$\widetilde{A}_{\mathbf{M},\lambda} \widetilde{A}_{\mathbf{M},\mu} - \rho\, \widetilde{A}_{\mathbf{M}',\lambda} \widetilde{A}_{\mathbf{M}',\mu} = \sum_{\nu} c^{\nu} \big(\widetilde{A}_{\mathbf{M},\nu} - \rho \widetilde{A}_{\mathbf{M}',\nu}\big) = t \sum_{\nu} c^{\nu} u_{\mathbf{M},\nu}.$$

Substituting $\widetilde{A}_{\mathbf{M},X} = \rho \widetilde{A}_{\mathbf{M}',X} + t\, u_{\mathbf{M},X}$ for $X = \lambda, \mu$ on the left,

$$\big(\rho \widetilde{A}_{\mathbf{M}',\lambda} + t u_{\mathbf{M},\lambda}\big)\big(\rho \widetilde{A}_{\mathbf{M}',\mu} + t u_{\mathbf{M},\mu}\big) - \rho \widetilde{A}_{\mathbf{M}',\lambda} \widetilde{A}_{\mathbf{M}',\mu} = \rho(\rho-1) \widetilde{A}_{\mathbf{M}',\lambda} \widetilde{A}_{\mathbf{M}',\mu} + t\big(\cdots\big),$$

with $(\cdots)$ the last two terms of (9). Divide by t. □

Theorem 13 (simplicity)**.** $\operatorname{ord}_{H_{ab}} c^{\nu}_{\lambda\mu} \geq -1$ *for all* λ, μ, ν *and all* $a < b$*: every pole of a structure constant on a long-root divisor is simple.*

Proof. Induction along $\prec$. If $\{a,b\} \not\subseteq \nu$ then $\operatorname{ord}_H \widetilde{A}_{\nu,\nu} = 0$ by (D), and (3) gives $\operatorname{ord}_H c^{\nu} \geq \min\big(0, \min_{\kappa \prec \nu} \operatorname{ord}_H c^{\kappa}\big) \geq -1$ by induction and (R).

If $\{a,b\} \subseteq \nu$, apply (9) with $\mathbf{M} = \nu$. Since $|\nu'| < |\nu|$, (T) gives $\widetilde{A}_{\nu',\nu} = 0$, so $u_{\nu,\nu} = \widetilde{A}_{\nu,\nu}/t$ and $\operatorname{ord}_H u_{\nu,\nu} = 1 - 1 = 0$: it is a unit of $\mathcal{O}$. Solving (9) for that term,

$$c^{\nu} = u_{\nu,\nu}^{-1}\Big(\frac{\rho(\rho-1) \widetilde{A}_{\nu',\lambda} \widetilde{A}_{\nu',\mu}}{t} + \big(\text{terms in } \mathcal{O}\big) - \sum_{\kappa \prec \nu} c^{\kappa} u_{\nu,\kappa}\Big),$$

where, as in Theorem 10, only κ with $\nu \geq \kappa$ and $\kappa \neq \nu$ contribute to the sum, since $u_{\nu,\kappa} = 0$ unless $\nu \geq \kappa$ by (T). Each $\mathrm{ord}_H(c^\kappa u_{\nu,\kappa}) \geq -1$ by induction and $u \in \mathcal{O}$, and the explicit term has $\mathrm{ord}_H = -1$ at worst. Hence $\mathrm{ord}_H\, c^\nu \geq -1$. □

Corollary 14 (the residue recursion)**.** *Write $\gamma_\nu := \big(t\, c^\nu_{\lambda\mu}\big)\big|_H \in \mathcal{K}$ for the residue, zero exactly when $c^\nu_{\lambda\mu}$ is regular on H (Theorem 13). Then for every $\mathbf{M} \supseteq \{a, b\}$,*

$$\sum_\nu \gamma_\nu\, u_{\mathbf{M},\nu}\big|_H = \rho(\rho-1)\, \widetilde{A}_{\mathbf{M}',\lambda}\big|_H\, \widetilde{A}_{\mathbf{M}',\mu}\big|_H. \tag{10}$$

In particular, taking $\mathbf{M} = \nu$ and using that $u_{\nu,\nu}|_H$ is nonzero,

$$\gamma_\nu = \frac{\rho(\rho-1)\widetilde{A}_{\nu',\lambda}|_H \widetilde{A}_{\nu',\mu}|_H - \sum_{\kappa\prec\nu} \gamma_\kappa u_{\nu,\kappa}|_H}{\big[t^{-1}\widetilde{A}_{\nu,\nu}\big]\big|_H}. \tag{11}$$

Proof. Reduce (9) at order t^{-1}, which is legitimate because every c^ν has $\mathrm{ord}_H \geq -1$ by Theorem 13 and every $u_{\mathbf{M},\nu}$ lies in $\mathcal{O}$. For (11), take $\mathbf{M} = \nu$ and isolate the $\kappa = \nu$ term as in the proof of Theorem 13. □

7 The created pole is real

Theorem 15 (Gap `nonvanishing`, on the minimal stratum)**.** *Fix $a < b$ and λ, μ, and let ν be* minimal *with respect to Motegi's order among the set*

$$\mathcal{S}(a,b) := \big\{\kappa \;:\; \kappa \geq \lambda,\ \kappa \geq \mu,\ \{a,b\} \subseteq \kappa,\ \kappa \setminus \{a,b\} \geq \lambda,\ \kappa \setminus \{a,b\} \geq \mu\big\},$$

and put $\nu' := \nu \setminus \{a, b\}$. If

$$\widetilde{A}_{\nu',\lambda}\big|_H \neq 0 \quad \textit{and} \quad \widetilde{A}_{\nu',\mu}\big|_H \neq 0, \tag{12}$$

then $\mathrm{ord}_{H_{ab}}\, c^\nu_{\lambda\mu} = -1$ exactly, with residue

$$\mathrm{Res}_{H_{ab}}\, c^\nu_{\lambda\mu} = \frac{\rho(\rho-1)\, \widetilde{A}_{\nu',\lambda}\big|_H\, \widetilde{A}_{\nu',\mu}\big|_H}{\big[t^{-1}\widetilde{A}_{\nu,\nu}\big]\big|_H} \;\neq 0. \tag{13}$$

Proof. By minimality, no $\kappa \prec \nu$ with $\kappa \geq \lambda, \mu$ lies in $\mathcal{S}(a,b)$, and the same is then true of every $\kappa' \leq \kappa$. So Theorem 10 applies at every such κ:

$$\mathrm{ord}_H\, c^\kappa_{\lambda\mu} \geq 0 \qquad \text{for all } \kappa \prec \nu. \tag{14}$$

There is no propagation to handle — this is the only place minimality is used, and it is exactly what Theorem 10 was proved for.

Now compute the numerator of (3) using Theorem 5 in the form $\widetilde{A}_{\nu,X} = \rho \widetilde{A}_{\nu',X} + t\, u_{\nu,X}$ with $u_{\nu,X} \in \mathcal{O}$:

$$\widetilde{A}_{\nu,\lambda}\widetilde{A}_{\nu,\mu} - \sum_{\kappa\prec\nu} c^\kappa \widetilde{A}_{\nu,\kappa} = \rho^2 \widetilde{A}_{\nu',\lambda}\widetilde{A}_{\nu',\mu} - \rho \sum_{\kappa\prec\nu} c^\kappa \widetilde{A}_{\nu',\kappa} \;+\; t\, E, \tag{15}$$

where $E = \rho\big(\widetilde{A}_{\nu',\lambda} u_{\nu,\mu} + \widetilde{A}_{\nu',\mu} u_{\nu,\lambda}\big) + t\, u_{\nu,\lambda} u_{\nu,\mu} - \sum_{\kappa\prec\nu} c^\kappa u_{\nu,\kappa}$ lies in $\mathcal{O}$, by (R), by $u \in \mathcal{O}$, and by (14).

The middle sum of (15) is the whole of (2) at $\mathbf{M} = \nu'$. Indeed $\widetilde{A}_{\nu',\kappa} = 0$ unless $\nu' \geq \kappa$ by (T); and $\nu' \geq \kappa$ forces $|\kappa| \leq |\nu'| < |\nu|$, hence $\kappa \neq \nu$, and $\nu \geq \kappa$ by Lemma 3(ii) applied to $\nu' \subseteq \nu$ — so $\kappa \prec \nu$ and every term of (2) at ν' already occurs in the sum. Therefore

$$\sum_{\kappa \prec \nu} c^{\kappa} \widetilde{A}_{\nu',\kappa} = \sum_{\text{all } \kappa} c^{\kappa} \widetilde{A}_{\nu',\kappa} = \widetilde{A}_{\nu',\lambda} \widetilde{A}_{\nu',\mu},$$

and (15) collapses to

$$\widetilde{A}_{\nu,\lambda} \widetilde{A}_{\nu,\mu} - \sum_{\kappa \prec \nu} c^{\kappa} \widetilde{A}_{\nu,\kappa} = \rho(\rho - 1)\, \widetilde{A}_{\nu',\lambda} \widetilde{A}_{\nu',\mu} + t\, E. \tag{16}$$

By Proposition 7, $\operatorname{ord}_H \rho = \operatorname{ord}_H(\rho - 1) = 0$; by (12), $\operatorname{ord}_H \widetilde{A}_{\nu',\lambda} = \operatorname{ord}_H \widetilde{A}_{\nu',\mu} = 0$. So the right-hand side of (16) has $\operatorname{ord}_H = 0$. As in Theorem 10, $\operatorname{ord}_H \widetilde{A}_{\nu,\nu} = 1$ exactly, so $\operatorname{ord}_H c^{\nu} = 0 - 1 = -1$, and dividing (16) by $t^{-1}\widetilde{A}_{\nu,\nu}$ and reducing gives (13). □

Remark 16 (what made it work). The sum over κ collapses to a *single* term before any cancellation can occur, because (2) at the smaller specialisation point ν' reassembles it. The session brief hoped for this through a residue of $\widetilde{A}$ on $N - 2$ variables; the object that actually does the work is the *restriction* of $\widetilde{A}$, and the residue taken is that of c, not of $\widetilde{A}$. The deletion is the same deletion twice over, as the brief guessed: Lemma `comb` of [Q488] deletes the two smallest elements of ν, and Theorem 5 deletes $\{a, b\}$ from the specialisation — cost 2, the size of a long root's support, in both.

Corollary 17 (the converse, qualitatively)**.** *If* $|\nu| - \max(|\lambda|, |\mu|) \geq 2$ *then, under* (12)*, the family* $\{c^{\kappa}_{\lambda\mu}\}_{\kappa}$ *is not contained in* $\mathbb{C}(q)[w^{\pm 1}]$*: taking* $\{a, b\}$ *to be the two smallest elements of* ν*, Lemma* `comb` *of [Q488] puts* $\nu \in \mathcal{S}(a, b)$*, so* $\mathcal{S}(a, b) \neq \emptyset$ *and any minimal element of it carries a simple pole on* H_{ab} *by Theorem 15.*

8 Scope: what is closed, and what is not

I state the limits of the above sharply, because they are larger than the "verified at $N \leq 3$" habit would suggest.

Gap 1 (`nonvanH`: hypothesis (12))**.** $\operatorname{ord}_{H_{ab}} \widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ *whenever* $\mathbf{M} \geq \mathbf{I}$ *and* $\{a, b\} \cap \mathbf{M} = \emptyset$.

This is exactly the *localisation at* H of the converse of triangularity ($\mathbf{M} \geq \mathbf{I} \Rightarrow \widetilde{A}_{\mathbf{M},\mathbf{I}} \neq 0$), which [Q488] already carries as measured (84/84 at $N \leq 3$) and not proved. It is not implied by that converse — a nonzero Laurent polynomial may be divisible by $w_a w_b - 1$ — so it is a genuinely separate hypothesis, and I mark it as such rather than quietly promoting a measurement. Status: measured 64/64 at $N = 3$ with a discriminating positive control (§9); by Theorem 5 the case $\{a, b\} \subseteq \mathbf{M}$ reduces to it, so the single statement above is the whole of what is needed. In the equal-cardinality case $|\mathbf{M}| = |\mathbf{I}|$ the branching sum has one term, so `nonvanH` reduces further to non-vanishing of the *type* A weight function $W_{N,r}(w_{\mathbf{M}}|\mathbf{w}|\mathbf{I})$ on H — a type A statement, on a divisor that is not of type A form.

Gap 2 (`propagation`: non-minimal ν)**.** *For* ν *not minimal in* $\mathcal{S}(a, b)$*,* (11) *gives*

$$\gamma_{\nu} \cdot \big[t^{-1}\widetilde{A}_{\nu,\nu}\big]\big|_H = \rho(\rho - 1)\widetilde{A}_{\nu',\lambda}\big|_H \widetilde{A}_{\nu',\mu}\big|_H - \sum_{\kappa \prec \nu} \gamma_{\kappa}\, u_{\nu,\kappa}\big|_H,$$

and what is unproved is exactly that the two terms on the right do not cancel.

This is a strict improvement on the previous formulation of Gap `nonvanishing` ("the absence of conspiratorial cancellation among the surviving **K**-terms of the branching formula"): the quantity that must not vanish is now written down, the sum is over structure constants rather than branch terms, and the only new analytic object is the first-order collapse deviation $u_{\nu,\kappa}|_H$ of (8).

How much is the minimal stratum? The minimality hypothesis of Theorem 15 is combinatorial, so it can be counted far beyond where the algebra is computable. Over all triples with $\nu \geq \lambda, \mu$ and jump ≥ 2, counting a triple as covered if *some* pair (a, b) makes ν minimal in $\mathcal{S}(a, b)$:

N	jump ≥ 2 triples	covered by Thm 15	%
2	1	1	100.0
3	13	12	92.3
4	127	99	78.0
5	1136	690	60.7
6	9748	4335	44.5

Coverage decreases with N. The single uncovered triple at $N = 3$ is $(\lambda, \mu, \nu) = (\emptyset, \emptyset, \{1, 2, 3\})$; at $N \geq 4$ the uncovered set is no longer confined to $\lambda = \mu = \emptyset$. So Theorem 15 is a partial closure of Gap `nonvanishing` on a stratum that is *asymptotically small*, and Gap 2 is the bulk of the problem, not an edge case. Reporting the $N \leq 3$ figures alone would have read as a closure; it is not one.

Type. Theorem 5 is verified in type B as well as C, with the same ρ (§9). Theorems 10, 13 and 15 use only (T), (R), (D), (S), (A) and Theorem 5, so they hold in type B to whatever extent those facts do; [Q488] records $w_a = -1$ as the place where type B could differ, and that is a short-root divisor, not a long-root one.

9 Computational verification, with controls and plants

All computation is pure Python + SymPy over $\mathbb{C}(q)(w_1, \ldots, w_N)$ (SageMath is not installed in this container). The engine is `q484_engine.py`, carried over unchanged, which evaluates $F_{N,n,r}(\mathbf{u}|\mathbf{w}|\mathbf{I})$ by brute-force summation over S_n from Motegi's symmetrisation formula. **It never uses the branching formula** (4), on which every proof above rests, so the tables below are independent corroboration rather than a second reading of one engine. Scripts and logs: `proofs/code-1016c2-nonvanishing/`.

The gate (`gate_q0.py`, `gate2.py`). $N = 2$: 20 nonzero constants, 1 pole record. $N = 3$: 115 nonzero constants, 25 pole records, 13 distinct triples. In 25/25 records: multiplicity 1, $\operatorname{ord}_q c = 1$, $\operatorname{ord}_q \operatorname{Res} = 1$. Hence $c|_{q=0} = 0$ in every one, which is the whole of Rick's M2 on these triples.

Theorem 5 (`collapse_test.py`), with two plants. Over all (a, b), all $\mathbf{M} \supseteq \{a, b\}$ and all $\mathbf{I}$:

	$N = 2$ C	$N = 3$ C	$N = 3$ B
pairs tested	4	48	48
(5) holds	4	48	48
of which *non-degenerate* (some side $\neq 0$ on H)	1	12	12
P1 $\rho \mapsto \rho(1 + q)$: failures	1	12	12
P2 $\rho \mapsto 1$ (untwisted): failures	1	12	12

The two plants sit in the same block — the multiplier — which is the block the theorem is about; each fails on *exactly* the non-degenerate sub-domain and nowhere else, which is the informative pattern. The non-degenerate count is reported because a pair with both sides zero on H cannot distinguish any multiplier: at $N = 2$ the domain has *one* item, so the $N = 2$ run alone would have been decoration. **P2** is the sharp one: $\rho = 1$ is the hypothesis that a long-root divisor is a collision divisor in the sense of [Q488] §5, and it fails.

Gap 1 (`nonvanH_test.py`), with a discriminating positive control. At $N = 3$ type C, over all (a, b) and all $\mathbf{M} \geq \mathbf{I}$:

domain	n	$\mathrm{ord}_H \widetilde{A}_{\mathbf{M},\mathbf{I}}$
$\{a,b\} \cap \mathbf{M} = \emptyset$ (the hypothesis (12) needs)	12	0 in 12/12
$\lvert\{a,b\} \cap \mathbf{M}\rvert = 1$	52	0 in 52/52
$\{a,b\} \subseteq \mathbf{M}$ (positive control)	41	0 in 12, ≥ 1 in 29

The control is the sharpness of (A): on $\{a, b\} \subseteq \mathbf{M}$ the prediction is $\mathrm{ord}_H \widetilde{A}_{\mathbf{M},\mathbf{I}} = 0 \iff \mathbf{M} \setminus \{a, b\} \geq \mathbf{I}$, and it agrees 41/41. That control's domain contains *both* values (12 zeros and 29 positives), so the scan demonstrably discriminates — a scan returning only one value over its whole domain would not have been evidence that it can detect a violation at all.

Theorem 15 (`residue_test.py`), with three plants. For every minimal ν, the measured $\mathrm{Res}_H c^{\nu}$ is compared against the closed form (13), whose three ingredients are evaluated separately. At $N = 2$: 1/1 match. At $N = 3$: see the log; every case computed matched, with no mismatch. The residues are visibly proportional to q, as Proposition 7 requires; e.g. $\mathrm{Res}_{H_{12}} c^{\{1,2\}}_{\emptyset\emptyset} = -q(q-1)^4$ at $N = 2$, and at $N = 3$

$$\mathrm{Res}_{H_{12}} c^{\{1,2\}}_{\emptyset\emptyset} = \frac{q(q-1)^4(qw_3 - w_1)(qw_3^2 - 1)(qw_1w_3 - 1)}{w_1}.$$

Pre-registered plants, in two different blocks: **P1** $\rho(\rho-1) \mapsto \rho(\rho-1)(1+q)$ and **P2** $\rho(\rho-1) \mapsto \rho^2$, both in the multiplier block, fire on every case; **P3** $\widetilde{A}_{\nu',\mu} \mapsto \widetilde{A}_{\nu',\lambda}$, in the label block, fires on every case with $\lambda \neq \mu$ and is *informatively spared* when $\lambda = \mu$, where it is the identity by construction. Naming which plant covers which block is the point: P1/P2 test the scalar, P3 tests that the two labels enter in the two slots claimed.

Coverage (`coverage.py`). The table of §8, computed combinatorially for $N \leq 6$. The minimisation is performed over the support *bound* $\kappa \geq \lambda, \mu$, a superset of the true support; more competitors make minimality harder, so the criterion is conservative and the covered counts are lower bounds.

Source hygiene

Every fact attributed to Motegi is from arXiv:2610.15344 and was read first-hand in the e-print source `typebcweightfunctions_arxivversionfinal-3.tex`, with line numbers recorded in [Q488]; the branching formula (4) is transcribed from L3203–L3236 there. Rick's M2 is from his review PDF at `mail/attachments/813/2026-10-16-review-clio-Q484-motegi.pdf`; I reproduced the measurement on my own engine rather than citing his numbers, and the reproduction used my engine's figures (20, 115 nonzero constants), which are not his counting convention (30, 195) — the *conclusion* is what was compared, not the totals. No claim above rests on a document whose author is also its subject.